

MINISTRY OF ENERGY & INFRASTRUCTURE
United Arab Emirates

TECHNICAL ASSESSMENT

MOEI HR Companion

Batch 2

*An HR product has two very different users: an employee who needs help,
and a manager who needs to understand their people.*

Build one side well — and show us you understand the other.

Duration: 2 Hours · Total Marks: 100

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1. The Brief

Over the next two hours you will build the MOEI HR Companion. Here is the idea in one breath: HR serves two people whose needs point in opposite directions.

- **The employee says “help me, quickly.”** Answer my question, check my balance, do this task for me — with less digging.
- **The manager says “help me understand and decide.”** Who on my team is doing well? Who might be about to leave? Who is ready to grow?

Those are two genuinely different products. So your task is not to build both. Pick one door, build it well, and in your README describe how you would build the other. Both doors look at the same employees — just from opposite ends.

What we are really evaluating. Your judgment as an AI developer, not the volume of code. Can you turn a messy human request into something grounded and trustworthy? Can you design a real product rather than a single prompt? A clean, finished one-door build plus a clear design of the other beats a broken attempt at everything.

2. The Two Doors — pick one

Door A — The employee assistant

An assistant that doesn't just answer HR questions (grounded in the official policy documents) but can act for the employee. Ideas, as inspiration only — use your own if you have a better one:

- Submit a request on the employee's behalf (e.g. leave, a training nomination) after checking the rules.
- Check “am I eligible for this?” against the employee's own record, not just the general policy.
- Walk someone through a process step by step, then complete the step for them.

The bar: an assistant that only answers questions — a plain FAQ bot — is the floor, not the goal. The interesting part is the doing.

Door B — The management view

A tool that turns employee data into insight a manager can act on. Ideas, again as inspiration only:

- Who is investing most in learning, and who has gone cold.
- Who looks ready for promotion (strong performance, recent learning, a while since their last move).
- Who might be a flight risk, or where a team has a skills gap.
- “Ask your workforce” — a manager asks a question in plain language and the tool reasons over the data and answers with the actual people and why.

The bar: a static dashboard of charts is the floor. Real analytics doesn't just show that something changed — it explains why and suggests what to do. Let the AI interpret, not just plot.

The one quality bar. Whichever door you choose, it must show real AI capability for that side — an employee door that acts, or a management view that interprets and recommends. The generic version (a pure FAQ bot, or charts with no interpretation) misses the point of the task.

3. Build One, Design the Other

Build one door as a working product. Then, in your README, spend a short section designing the door you did not build: what it would do, what data and AI it would need, and why that user (employee or manager) would care. We are looking for evidence that you understand the whole product, even though you only had time to build half of it. A thoughtful page here counts for real marks (see Section 8).

4. The Data You'll Work With

You get two grounding sources, and they do different jobs:

- **The HR knowledge base (policy).** Official MOEI / UAE Federal HR documents in English and Arabic — the HR Law, performance and training systems, code of ethics, and related guides. Ground policy answers here. Don't invent rules, numbers, or entitlements.
- **The synthetic employee dataset (people).** A spreadsheet of ~260 fictional employees across six sheets — Employees, Performance, Training, Leave, Movement, plus a Read Me that explains every column. This is the spine both doors share: the employee door reads and writes one person's record; the management door reasons across everyone's. Nothing is pre-computed — things like tenure or promotion-readiness are yours to derive.

5. New to HR? Start Here

Most of you are students, and that's fine — you don't need HR experience to do well. Here are the only terms you really need:

Term	In plain words
Performance cycle	The yearly review where a manager scores how an employee did against their goals.
High-potential / promotion-ready	Someone who looks ready to take on a bigger role soon — usually strong performance plus recent learning.
Attrition / flight risk	"Attrition" is people leaving. "Flight risk" is someone who looks likely to leave soon.
Headcount	How many people work somewhere — e.g. headcount per department.
Skills gap	A skill a team needs but doesn't have enough of yet.

Ask us. If you're unsure what a manager or employee would actually want, ask — that's a fair question, and asking beats guessing. Understanding the user is part of the job.

6. How You'll Be Graded

There's no hidden scheme to game — the more you build well, the better you score.

What we grade	What good looks like	Marks
The door you built	It works, and it's a real product — not the generic version. The core challenge for that door is solved (an employee door that acts; a management view that interprets and recommends).	40
AI engineering	A current model with a real reason; one deliberate agentic / structured pattern; graceful failure handling.	20
The door you didn't build	A clear, credible design of the other side: what it does, what data and AI it needs, why that user cares.	15
Bilingual (EN / AR)	Works in both, replies in the language used; Arabic handled properly, not bolted on.	10
Correctness & judgment	Policy answers grounded in the KB / people-insights honest with the data; runs from a clean clone; thoughtful REFLECTION.	15
Total		100

Quality gate. The door you build must show real AI for its side. A pure FAQ bot (employee), or a static dashboard with no interpretation (management), caps the 40-point block at about 15 — however polished it looks.

7. Rules, Requirements & Submission

Item	Detail
Duration	Two hours for code (small leeway to save and push). The ~2-minute walkthrough video is recorded afterwards, with an extra 10–15 minutes.
Work mode	Individual work. Don't share code, prompts, or solutions during the window, and don't have someone else build your solution.
Tools	Any AI assistant or coding agent, any language or framework, the provided knowledge base and dataset, and your OpenRouter key. The open web is fine for docs and debugging.
AI policy	Use AI heavily — that's the point, and there's no penalty for AI-generated code. But decide your own design before you start: pasting this brief into a model and shipping the result is the fastest route to an average, identical-looking submission.

Your submission must include

- **Working code for one door** (employee or management), grounded in the right source — policy answers in the knowledge base, people-insights in the dataset — and failing gracefully when the API errors or an answer isn't there.
- **Both languages** — English and Arabic, replying in the language the user writes in.
- **A README** covering: how to run it; your model choice and why; one line on your deliberate AI pattern (structured output, tool call, or multi-step); your design of the door you didn't build; and the features you completed.
- **A completed REFLECTION.md** (template provided) at the repository root.
- **Your GitHub URL** — with the API key read from the environment, never committed.

Submit your code, REFLECTION.md and repo URL to Google Classroom before the 2-hour hard stop, then record your walkthrough separately afterwards — show the door you built, demo it, and mention how you'd build the other.

*Build something an employee — or their manager — would actually thank you for.
Good luck.*

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